
PIXO UI

Step-by-Step Implementation Guide

Where Ideas Find Voice

A comprehensive walkthrough for building a modern, dark-themed blog platform with glass morphism navigation, scroll animations, dynamic filtering, search overlays, and responsive mobile design.

Technologies: HTML5, CSS3 (Custom Properties, Flexbox, Grid), Tailwind CSS, Vanilla JavaScript, Intersection Observer API, Font Awesome

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1. Project Overview & Architecture

1.1 What We Are Building

PIXO is a modern, dark-themed blog platform that combines elegant typography with immersive visual design. The interface features a deep charcoal background (#0a0a0a) punctuated by a warm amber accent color (#e8a023), creating a sophisticated reading environment that draws the eye to content. The design philosophy centers on the principle that great UI should feel invisible: users should be immersed in the content, not distracted by chrome.

The platform consists of several key sections: a fixed glass-morphism navigation bar that responds to scroll position, a full-height hero section with floating animations, a trending marquee ticker, category filter pills, a responsive blog post grid, an editor's pick feature, a newsletter subscription section, and a comprehensive footer. Each section is enriched with scroll-reveal animations, hover interactions, and responsive breakpoints that ensure a polished experience across all device sizes.

1.2 Technology Stack

Technology	Purpose	Version / CDN
HTML5	Semantic page structure	Living Standard
CSS Custom Properties	Design tokens & theming	CSS3
Tailwind CSS	Utility-first styling	v3.4 (CDN)
Font Awesome	Icon library	v6.5.1 (CDN)
Space Grotesk	Heading typeface	Google Fonts
DM Sans	Body typeface	Google Fonts
Vanilla JavaScript	Interactivity & DOM manipulation	ES6+
Intersection Observer	Scroll-based animations	Web API

1.3 File Structure

The entire PIXO interface is implemented as a single HTML file, making it easy to understand and deploy. The code is organized into three main sections: the <head> containing meta tags, font imports, Tailwind configuration, and custom CSS; the <body> containing all semantic HTML markup for each UI section; and a <script> block at the bottom containing all JavaScript logic. For production, you would typically split these into separate files (styles.css, app.js, components/), but the single-file approach is ideal for learning and prototyping.

```
index.html |-- <head> | |-- Meta tags & title | |-- Google Fonts (Space Grotesk, DM Sans) | |-- Font Awesome 6.5.1 | |-- Tailwind CSS CDN + config | |-- Custom CSS
```

```
(variables, animations, components) |-- <body> | |-- Reading progress bar | |-- Toast  
notification container | |-- Navigation (desktop + mobile) | |-- Search overlay | |--  
Hero section | |-- Trending marquee | |-- Categories section | |-- Blog posts grid | |--  
Editor's pick | |-- Newsletter | |-- Footer |-- <script> |-- Post data array |-- Render  
functions |-- Category filter logic |-- Scroll reveal observer |-- Nav scroll glass  
effect |-- Reading progress |-- Search overlay logic |-- Mobile menu toggle |--  
Newsletter form |-- Toast utility
```

2. Step 1: Foundation - HTML Structure & CSS Variables

2.1 CSS Custom Properties (Design Tokens)

The design system is anchored by a set of CSS custom properties defined in the `:root` selector. These variables serve as design tokens that ensure consistency across the entire interface. By centralizing color values, we enable easy theming and maintenance. If you ever need to switch to a light theme or a different accent color, you only need to change these variables in one place. This approach is far superior to hardcoding hex values throughout the stylesheet.

```
:root { --bg: #0a0a0a; /* Deep charcoal background */ --fg: #ecec; /* Off-white foreground text */ --muted: #7a7a7a; /* Gray for secondary text */ --accent: #e8a023; /* Warm amber accent */ --accent-glow: rgba(232, 160, 35, 0.12); /* Accent with low opacity */ --card: #141414; /* Slightly lighter than bg for cards */ --border: #222222; /* Subtle border color */ --surface: #1a1a1a; /* Elevated surface color */ }
```

The color palette follows a deliberate hierarchy: `--bg` is the deepest layer (page background), `--surface` is one step lighter (used for elevated elements like the stats card and search input), `--card` is another step (blog post cards), and `--border` sits at the edge of visibility. The `--accent` color (`#e8a023`) is a warm amber that provides excellent contrast against the dark background (WCAG ratio of approximately 8.5:1 for large text), while `--accent-glow` at 12% opacity creates subtle highlight backgrounds without overwhelming the dark aesthetic.

2.2 Global Reset & Scrollbar Styling

A universal reset removes default browser margins and padding, ensuring consistent layout across browsers. The scrollbar is styled to match the dark theme: a thin 6px track with dark colors that blends seamlessly into the background. Smooth scroll behavior is enabled globally so that anchor navigation (e.g., clicking nav links) animates smoothly rather than jumping. The body is set to `overflow-x: hidden` to prevent horizontal scroll from decorative elements that may extend beyond the viewport.

```
* { margin: 0; padding: 0; box-sizing: border-box; } html { scroll-behavior: smooth; scrollbar-width: thin; scrollbar-color: var(--border) var(--bg); } body { font-family: 'DM Sans', sans-serif; background: var(--bg); color: var(--fg); overflow-x: hidden; opacity: 0; animation: bodyFadeIn 0.6s ease forwards; } @keyframes bodyFadeIn { to { opacity: 1; } } /* Custom Scrollbar for WebKit browsers */ ::-webkit-scrollbar { width: 6px; } ::-webkit-scrollbar-track { background: var(--bg); } ::-webkit-scrollbar-thumb { background: var(--border); border-radius: 3px; }
```

The `bodyFadeIn` animation is a subtle but important detail: the page starts at opacity 0 and fades in over 0.6 seconds. This prevents the flash of unstyled content (FOUC) that can occur when custom fonts are still loading. Combined with the `font-display: swap` behavior of Google Fonts, this creates a polished loading experience where the user never sees a jarring transition from fallback to custom fonts.

2.3 Tailwind CSS Configuration

Tailwind CSS is loaded via CDN and configured to extend the default theme with custom font families and color mappings. The `fontFamily` extension registers Space Grotesk as the heading font and DM Sans as the body font, making them available via `font-heading` and `font-body` utility classes. The `colors` extension maps Tailwind's color system to our CSS custom properties, allowing us to use utilities like `bg-bg`, `text-fg`, `border-border`, and `text-accent`.

```
tailwind.config = { theme: { extend: { fontFamily: { heading: ['"Space Grotesk"',  
'sans-serif'], body: ['"DM Sans"', 'sans-serif'], }, colors: { bg: 'var(--bg)', fg:  
'var(--fg)', muted: 'var(--muted)', accent: 'var(--accent)', card: 'var(--card)',  
border: 'var(--border)', surface: 'var(--surface)', } } } }
```

3. Step 2: Navigation Bar with Glass Morphism

The navigation bar is one of the most visually distinctive elements of the PIXO interface. It employs a glass morphism effect where the background is semi-transparent with a backdrop blur, allowing the page content to subtly show through. When the user scrolls past 50 pixels, the navigation transitions to a more opaque state with a visible border, creating a clear visual distinction between the "at rest" and "scrolled" states. The nav is fixed to the top of the viewport with a z-index of 1000, ensuring it stays above all content.

3.1 Desktop Navigation HTML

The navigation structure is divided into three zones: the logo on the left, the center navigation links (hidden on mobile), and the right-side action buttons (search and write CTA). The logo uses the Space Grotesk font at 2xl size with the accent color for "PIXO" and foreground color for the trailing period, creating a memorable brand mark.

```
<nav class="nav-glass fixed top-0 left-0 w-full z-[1000]" id="mainNav"
role="navigation" aria-label="Main navigation"> <div class="max-w-7xl mx-auto px-6 py-4
flex items-center justify-between"> <!-- Logo --> <a href="#" class="font-heading
text-2xl font-bold tracking-tight" style="color: var(--accent);"> PIXO<span
style="color: var(--fg);">.</span> </a> <!-- Desktop Links --> <div class="hidden
md:flex items-center gap-8"> <a href="#hero" class="...">Home</a> <a href="#posts"
class="...">Explore</a> <a href="#categories" class="...">Categories</a> <a
href="#newsletter" class="...">Newsletter</a> </div> <!-- Right Actions --> <div
class="hidden md:flex items-center gap-4"> <button id="searchToggle" class="..."
aria-label="Search"> <i class="fa-solid fa-magnifying-glass ..."></i> </button> <a
href="#" class="btn-accent px-5 py-2 rounded-full ..."> Write <i class="fa-solid
fa-pen-nib text-xs"></i> </a> </div> <!-- Mobile Hamburger --> <button
id="mobileToggle" class="md:hidden ..." aria-label="Open menu"> <i class="fa-solid
fa-bars text-[var(--fg)]"></i> </button> </div> </nav>
```

3.2 Glass Morphism CSS

Glass morphism is achieved through a combination of semi-transparent background and backdrop-filter: blur(). The initial state uses rgba(10, 10, 10, 0.6) with a 20px blur, creating a frosted glass effect. The border-bottom starts as transparent, then becomes visible when scrolled. This dual-state approach ensures the nav looks clean when the page first loads (at the top) and provides a clear visual boundary once the user starts scrolling.

```
.nav-glass { background: rgba(10, 10, 10, 0.6); backdrop-filter: blur(20px);
-webkit-backdrop-filter: blur(20px); border-bottom: 1px solid transparent; transition:
all 0.4s ease; } .nav-glass.scrolled { background: rgba(10, 10, 10, 0.85);
border-bottom-color: var(--border); }
```

3.3 Scroll-Aware Background Change

The JavaScript scroll listener toggles the "scrolled" class based on the scroll position. Using a threshold of 50 pixels prevents the class from flickering during tiny scroll events. The { passive: true

} option improves scroll performance by telling the browser this listener will not call preventDefault().

```
const mainNav = document.getElementById('mainNav'); window.addEventListener('scroll',  
  () => { if (window.scrollY > 50) { mainNav.classList.add('scrolled'); } else {  
    mainNav.classList.remove('scrolled'); } }, { passive: true });
```

3.4 Mobile Hamburger Menu

On mobile devices (below the md breakpoint of 768px), the desktop links and actions are hidden, and a hamburger button is displayed instead. Tapping it reveals a full-screen overlay menu that slides in from the right using CSS transform. The overlay contains large, touch-friendly navigation links centered vertically. When a link is tapped, the menu automatically closes to provide a smooth user experience. The CSS transition uses a cubic-bezier curve (0.16, 1, 0.3, 1) for a satisfying spring-like animation.

```
/* Mobile Menu CSS */ .mobile-menu { transform: translateX(100%); transition: transform  
0.4s cubic-bezier(0.16, 1, 0.3, 1); } .mobile-menu.open { transform: translateX(0); }  
/* Mobile Menu JavaScript */ const mobileToggle =  
document.getElementById('mobileToggle'); const mobileMenu =  
document.getElementById('mobileMenu'); const mobileClose =  
document.getElementById('mobileClose'); mobileToggle.addEventListener('click', () =>  
mobileMenu.classList.add('open')); mobileClose.addEventListener('click', () =>  
mobileMenu.classList.remove('open')); // Close menu when any link is clicked  
document.querySelectorAll('.mobile-link').forEach(link => {  
link.addEventListener('click', () => mobileMenu.classList.remove('open')); });
```

4. Step 3: Hero Section with Floating Animation

The hero section is the first thing users see after the page loads, and it sets the visual tone for the entire experience. It spans the full viewport height (min-h-screen) and uses a two-column grid layout on large screens: the left column contains the featured article content (tag, headline, description, author, and CTA buttons), while the right column displays a large hero image with a floating stats card overlay. Decorative radial gradient glows in the background create depth and atmosphere without distracting from the content.

4.1 Hero Layout Structure

The hero uses a CSS Grid layout with `lg:grid-cols-2`, creating two equal-width columns on desktop. On smaller screens, the columns stack vertically with a gap of 12-16 units. The left column is wrapped in a "reveal" div for scroll animation, and the right column has a transition-delay of 0.15s so it animates slightly after the left content, creating a staggered entrance effect.

```
<header id="hero" class="relative min-h-screen flex items-center pt-24 pb-16
overflow-hidden"> <!-- Background Glow --> <div class="absolute top-1/2 right-1/4
w-[600px] h-[600px] rounded-full glow-pulse pointer-events-none" style="background:
radial-gradient(circle, rgba(232,160,35,0.12) 0%, transparent 70%);"> </div> <div
class="max-w-7xl mx-auto px-6 w-full"> <div class="grid lg:grid-cols-2 gap-12 lg:gap-16
items-center"> <!-- Left: Featured content --> <div class="reveal">...</div> <!--
Right: Hero image --> <div class="reveal relative" style="transition-delay: 0.15s;">
... </div> </div> </div> </header>
```

4.2 Background Glow Effects

Two radial gradient elements create ambient glow in the hero background. The primary glow is positioned at the vertical center and 25% from the right, using the accent color at 12% opacity. It fades from the center outward to transparent over 70% of its radius. This glow pulses gently using the `glowPulse` keyframe animation, which oscillates opacity between 12% and 20% while slightly scaling from 100% to 105%. The combination of low opacity and subtle scaling creates a breathing effect that feels organic rather than mechanical. A secondary, dimmer glow (5% opacity) sits at the bottom left for visual balance.

```
@keyframes glowPulse { 0%, 100% { opacity: 0.12; transform: scale(1); } 50% { opacity:
0.2; transform: scale(1.05); } } .glow-pulse { animation: glowPulse 4s ease-in-out
infinite; }
```

4.3 Floating Stats Card

One of the hero's most charming details is the floating stats card positioned at the bottom-left of the hero image. It's absolutely positioned at `-bottom-6` and `-left-6`, overlapping the image border to create a layered, dimensional effect. The card uses the `--surface` background with a border, backdrop blur, and a deep box shadow. Inside, a fire icon in an accent-glow circle sits beside the "2.4K Active readers" text. The entire hero image container uses the `heroFloat` animation, which

gently bobs the image up and down by 12 pixels over a 5-second cycle, creating a sense of lightness and movement.

```
@keyframes heroFloat { 0%, 100% { transform: translateY(0px); } 50% { transform: translateY(-12px); } } .hero-float { animation: heroFloat 5s ease-in-out infinite; }  

```

5. Step 4: Trending Marquee

The trending marquee is a horizontally scrolling ticker that displays popular topics between the hero and the main content area. It uses a pure CSS animation to create an infinite horizontal scroll effect without any JavaScript. The key technique is content duplication: the marquee items are rendered twice within the same flex container, and the animation translates the entire track by -50%, creating a seamless loop when the first set scrolls out of view and the duplicate set takes its place.

```
<!-- Marquee HTML --> <div class="border-y border-[var(--border)] py-3 overflow-hidden
relative"> <div class="marquee-track flex items-center gap-8 whitespace-nowrap"
style="width: max-content;"> <span class="font-heading text-sm font-semibold"
style="color: var(--accent);">TRENDING</span> <span class="text-sm" style="color:
var(--muted);"> <i class="fa-solid fa-circle text-[4px] mx-2" style="color:
var(--accent);"></i> Design Systems at Scale </span> <!-- ... more items ... --> <!--
Duplicate set for seamless loop --> <span class="font-heading text-sm font-semibold"
style="color: var(--accent);">TRENDING</span> <!-- ... same items again ... --> </div>
</div> /* Marquee CSS */ @keyframes marquee { 0% { transform: translateX(0); } 100% {
transform: translateX(-50%); } } .marquee-track { animation: marquee 30s linear
infinite; }
```

The overflow-hidden on the parent container ensures only the visible portion of the marquee is shown. The 30-second animation duration creates a slow, readable scroll speed. The linear timing function ensures constant velocity without acceleration or deceleration, which is essential for a smooth infinite loop. Each topic is separated by a tiny accent-colored dot (4px circle) that acts as a visual separator. The "TRENDING" label appears in the accent color with font-semibold weight, making it stand out from the topic names rendered in muted gray.

6. Step 5: Categories Filter Pills

6.1 Pill Button HTML

The category pills are a set of horizontally scrollable buttons that allow users to filter blog posts by topic. Each pill is a button element with a border, rounded-full shape, and the `cat-pill` class. The "All Posts" pill starts with the `active` class, which gives it the accent background and dark text. The `data-cat` attribute stores the category identifier used by the JavaScript filter logic. The pills are wrapped in a flex container with `gap-3` and `flex-wrap`, so they naturally wrap to the next line on smaller screens.

```
<div class="flex flex-wrap gap-3" id="categoryContainer"> <button class="cat-pill active px-5 py-2 rounded-full text-sm font-heading border border-[var(--border)]" data-cat="all">All Posts</button> <button class="cat-pill px-5 py-2 rounded-full text-sm font-heading border border-[var(--border)]" style="color: var(--muted);" data-cat="technology">Technology</button> <!-- ... more pills ... --> </div>
```

6.2 Active State & Filter Logic

The CSS for `cat-pill` defines the hover and active states. When active or hovered, the pill background becomes the accent color, text turns dark, border matches the accent, and a subtle `translateY(-1px)` lift is applied. The JavaScript click handler uses event delegation on the `categoryContainer`: when a pill is clicked, all pills are deactivated, the clicked pill is activated, and the posts grid is re-rendered with the filtered category.

```
/* Category Pill CSS */ .cat-pill { transition: all 0.3s cubic-bezier(0.16, 1, 0.3, 1); cursor: pointer; white-space: nowrap; } .cat-pill:hover, .cat-pill.active { background: var(--accent); color: #0a0a0a; border-color: var(--accent); transform: translateY(-1px); } /* Category Filter JavaScript */ const categoryContainer = document.getElementById('categoryContainer'); categoryContainer.addEventListener('click', (e) => { const pill = e.target.closest('.cat-pill'); if (!pill) return; // Deactivate all pills document.querySelectorAll('.cat-pill').forEach(p => { p.classList.remove('active'); p.style.color = 'var(--muted)'; }); // Activate clicked pill pill.classList.add('active'); pill.style.color = ''; // Re-render posts with selected category activeCategory = pill.dataset.cat; renderPosts(activeCategory); });
```

7. Step 6: Blog Posts Grid

7.1 Post Card HTML Structure

Each blog post card is an article element with a fixed-height image container (200px), followed by a content section with padding. The card displays the category tag, read time, title (limited to 2 lines with line-clamp-2), excerpt (also 2 lines), and author information with avatar. The card uses the `--card` background and `--border` border, creating a subtle elevation from the page background. The `overflow-hidden` on the image container ensures the image zoom animation (on hover) doesn't bleed outside the card boundaries.

```
<article class="post-card reveal rounded-2xl border border-[var(--border)]
overflow-hidden" style="background: var(--card);"> <!-- Image --> <div
class="overflow-hidden" style="height: 200px;">  </div> <!-- Content -->
<div class="p-5"> <div class="flex items-center gap-2 mb-3"> <span class="px-2.5 py-0.5
rounded-full text-[10px] font-heading font-semibold uppercase tracking-wider"
style="background: var(--accent-glow); color: var(--accent);"> technology </span> <span
class="text-[11px]" style="color: var(--muted);">6 min</span> </div> <h3
class="font-heading text-base font-bold leading-snug mb-2 line-clamp-2" style="color:
var(--fg);"> Rethinking Component Architecture </h3> <p class="font-body text-sm
leading-relaxed mb-4 line-clamp-2" style="color: var(--muted);"> How modular design
patterns transform... </p> <div class="flex items-center gap-2.5">  <div> <p class="text-xs
font-medium" style="color: var(--fg);">Kavya Iyer</p> <p class="text-[10px]"
style="color: var(--muted);">Dec 15, 2024</p> </div> </div> </div> </article>
```

7.2 Card Hover Animations

The post card has two layers of hover interaction. First, the entire card lifts up by 6 pixels with an enhanced box shadow, creating a floating effect. Second, the image inside scales up by 6%, giving a subtle zoom effect that draws attention. Both transitions use the cubic-bezier (0.16, 1, 0.3, 1) curve, which creates a fast start with a gentle deceleration, feeling responsive yet smooth. The different durations (0.35s for the card lift, 0.5s for the image zoom) create a layered, organic feel.

```
.post-card { transition: transform 0.35s cubic-bezier(0.16, 1, 0.3, 1), box-shadow
0.35s ease; } .post-card:hover { transform: translateY(-6px); box-shadow: 0 20px 50px
rgba(0,0,0,0.4); } .post-card .card-img { transition: transform 0.5s cubic-bezier(0.16,
1, 0.3, 1); } .post-card:hover .card-img { transform: scale(1.06); }
```

7.3 Dynamic Rendering with JavaScript

Rather than hardcoding each card in HTML, the posts are defined as a JavaScript array of objects and rendered dynamically. The `renderPosts()` function filters the array based on the active category, clears the grid, creates article elements for each post, and appends them to the DOM. Each card receives a staggered transition-delay based on its index ($i * 0.08s$), creating a cascading reveal animation. After rendering, the `IntersectionObserver` is re-initialized to observe the new elements.

```
const posts = [ { id: 1, title: "Rethinking Component Architecture for Scale", excerpt:
"How modular design patterns and atomic methodology transform maintainability...",
category: "technology", author: "Kavya Iyer", authorImg:
"https://picsum.photos/seed/a1/48/48.jpg", date: "Dec 15, 2024", readTime: "6 min",
img: "https://picsum.photos/seed/post1/600/400.jpg" }, // ... 8 more post objects ];
function renderPosts(category) { const filtered = category === 'all' ? posts :
posts.filter(p => p.category === category); postsGrid.innerHTML = '';
filtered.forEach((post, i) => { const card = document.createElement('article');
card.className = 'post-card reveal rounded-2xl'; card.style.cssText = `background:
var(--card); transition-delay: ${i * 0.08}s;`; card.innerHTML = `...`;
postsGrid.appendChild(card); }); observeRevealElements(); }
```

The Editor's Pick section breaks the grid pattern with a wide, horizontal card that spans the full content width. It uses a 5-column grid (`lg:grid-cols-5`) where the image occupies 2 columns and the content occupies 3, creating an asymmetric layout that feels editorial and premium. The image has a hover zoom effect (scale 1.05 over 700ms), and the content side is vertically centered using flex with justify-center. The "Editor's Pick" badge is a filled accent pill with a star icon, distinguishing it from the category tags on regular post cards.

```
<section class="pb-24"> <div class="max-w-7xl mx-auto px-6"> <div class="reveal rounded-2xl border border-[var(--border)] overflow-hidden" style="background: var(--card);"> <div class="grid lg:grid-cols-5"> <!-- Image Side (2 cols) --> <div class="lg:col-span-2 overflow-hidden">  </div> <!-- Content Side (3 cols) --> <div class="lg:col-span-3 p-8 lg:p-12 flex flex-col justify-center"> <div class="flex items-center gap-3 mb-4"> <span class="inline-flex items-center gap-1.5 px-3 py-1 rounded-full text-xs font-heading font-semibold" style="background: var(--accent); color: #0a0a0a;"> <i class="fa-solid fa-star text-[8px]"></i> Editor's Pick </span> </div> <h2 class="font-heading text-2xl sm:text-3xl lg:text-4xl font-bold leading-tight mb-4" style="color: var(--fg);"> Why the Best Interfaces Are the Ones You Never Notice </h2> <p class="font-body text-base leading-relaxed mb-6" style="color: var(--muted);"> Great design disappears... </p> <!-- Author + Read CTA --> <div class="flex items-center justify-between"> <!-- Author info --> <!-- Read button --> </div> </div> </div> </div> </div> </div> </div>
```

9. Step 8: Newsletter Section

The newsletter section serves as both a content break and a conversion point. It uses the noise-overlay class to add a subtle texture via an SVG-based fractal noise filter at 3% opacity, creating a tactile, paper-like quality. A radial gradient in the background centers a warm amber glow at 6% opacity, providing a subtle spotlight effect. The content is centered in a max-w-2xl container with relative z-10 to sit above the noise overlay (z-index 0).

The form uses a flex layout that switches from column (mobile) to row (desktop) orientation. The email input has a custom focus state that applies an accent border and a 3px accent glow ring (box-shadow: 0 0 0 3px rgba(232, 160, 35, 0.15)). The submit button uses the btn-accent class with a rounded-xl shape. When submitted, the form prevents default behavior, validates the email, shows a toast notification, and clears the input.

```
<!-- Noise Overlay CSS --> .noise-overlay::before { content: ''; position: absolute;
inset: 0; opacity: 0.03; background-image: url("data:image/svg+xml, %3Csvg viewBox='0 0
256 256' xmlns='http://www.w3.org/2000/svg'%3E %3Cfilter id='n'%3E %3CfeTurbulence
type='fractalNoise' baseFrequency='0.85' numOctaves='4' stitchTiles='stitch'/%3E
%3C/filter%3E %3Crect width='100%25' height='100%25' filter='url(%23n)'/%3E
%3C/svg%3E"); pointer-events: none; z-index: 0; } <!-- Input Focus CSS -->
.email-input:focus { outline: none; border-color: var(--accent); box-shadow: 0 0 0 3px
rgba(232, 160, 35, 0.15); } <!-- Newsletter JavaScript --> const newsletterForm =
document.getElementById('newsletterForm'); newsletterForm.addEventListener('submit',
(e) => { e.preventDefault(); const email = document.getElementById('emailInput').value;
if (email) { showToast(`Welcome aboard! We'll send updates to ${email}`);
document.getElementById('emailInput').value = ''; } });
```

10. Step 9: Footer Component

The footer uses a 4-column grid layout (`lg:grid-cols-4`) with the brand column spanning 1 column and three link columns for Platform, Categories, and Company. Each link column has an uppercase, letter-spaced heading (`tracking-[0.15em]`) in the foreground color, followed by a spaced list of links in muted color that transition to accent on hover. The brand column includes the PIXO logo, a short description, and three social media icon links (X/Twitter, GitHub, Instagram) styled as small bordered squares that highlight with the accent color on hover.

Below the main grid, a bottom bar with a top border contains two pieces of micro-copy: a copyright notice and a tagline. These are centered on mobile and space-between on larger screens using `flex-col sm:flex-row`. The consistently small text size (`text-xs`) and muted color ensure the footer feels receded and doesn't compete with the main content above.

```
<footer class="border-t border-[var(--border)] pt-16 pb-8"> <div class="max-w-7xl
mx-auto px-6"> <div class="grid sm:grid-cols-2 lg:grid-cols-4 gap-10 mb-14"> <!-- Brand
Column --> <div class="lg:col-span-1"> <a href="#" class="font-heading text-2xl
font-bold" style="color: var(--accent);"> PIXO<span style="color: var(--fg);">.</span>
</a> <p class="font-body text-sm mt-4 leading-relaxed" style="color: var(--muted);"> A
modern platform for thinkers, creators, and storytellers. </p> <!-- Social icons -->
<div class="flex gap-3 mt-5"> <a href="#" class="w-9 h-9 rounded-lg border
border-[var(--border)] flex items-center justify-center hover:border-[var(--accent)]
hover:bg-[var(--accent-glow)] transition-all" aria-label="Twitter"> <i class="fa-brands
fa-x-twitter text-xs" style="color: var(--muted);"></i> </a> <!-- GitHub, Instagram...
--> </div> </div> <!-- Platform, Categories, Company link columns follow same pattern
--> </div> <!-- Bottom Bar --> <div class="border-t border-[var(--border)] pt-6 flex
flex-col sm:flex-row items-center justify-between gap-4"> <p class="text-xs"
style="color: var(--muted);"> 2024 PIXO. Crafted with intention. </p> <p
class="text-xs" style="color: var(--muted);"> Made for writers who care about their
craft. </p> </div> </div> </footer>
```

11. Step 10: Search Overlay with Keyboard Shortcuts

The search overlay is a full-screen modal that appears when the search button is clicked or when the Ctrl+K / Cmd+K keyboard shortcut is pressed. It uses a semi-transparent dark background (rgba(10,10,10,0.92)) with backdrop-blur-xl for a deep frosted glass effect. The search input is centered at the top of the screen (pt-32) in a max-w-2xl container, with a search icon on the left and a close button on the right. Results appear below the input in real-time as the user types.

```
<!-- Search Overlay HTML --> <div id="searchOverlay" class="fixed inset-0 z-[1001]
bg-[rgba(10,10,10,0.92)] backdrop-blur-xl flex items-start justify-center pt-32 px-6
hidden" role="dialog" aria-label="Search"> <div class="w-full max-w-2xl"> <div
class="relative"> <i class="fa-solid fa-magnifying-glass absolute left-5 top-1/2
-translate-y-1/2 text-[var(--muted)]"></i> <input id="searchInput" type="text"
placeholder="Search articles, topics, authors..." class="w-full bg-[var(--surface)]
border border-[var(--border)] rounded-2xl py-4 pl-14 pr-14 text-lg text-[var(--fg)]
font-body email-input" autofocus> <button id="searchClose" class="absolute right-4
top-1/2 -translate-y-1/2 w-8 h-8 rounded-lg bg-[var(--border)] flex items-center
justify-center" aria-label="Close search"> <i class="fa-solid fa-xmark text-xs
text-[var(--fg)]"></i> </button> </div> <div id="searchResults" class="mt-4
space-y-2"></div> <p class="text-center text-[var(--muted)] text-sm mt-6">Press ESC to
close</p> </div> </div>
```

The search logic filters posts by title, excerpt, category, and author name using case-insensitive matching. Results are displayed as link elements with a thumbnail image, title, author, read time, and category badge. The overlay can be closed via the close button, the ESC key, or by clicking outside. The autofocus attribute on the input ensures the cursor is ready immediately when the overlay opens, and a setTimeout wrapper ensures the focus call happens after the display change.

```
// Open search on Ctrl+K / Cmd+K document.addEventListener('keydown', (e) => { if (e.key
=== 'Escape') { searchOverlay.classList.add('hidden'); searchInput.value = '';
searchResults.innerHTML = ''; } if ((e.metaKey || e.ctrlKey) && e.key === 'k') {
e.preventDefault(); searchOverlay.classList.remove('hidden'); setTimeout(() =>
searchInput.focus(), 100); } }); // Live search filtering
searchInput.addEventListener('input', () => { const q =
searchInput.value.toLowerCase().trim(); searchResults.innerHTML = ''; if (!q) return;
const results = posts.filter(p => p.title.toLowerCase().includes(q) ||
p.excerpt.toLowerCase().includes(q) || p.category.toLowerCase().includes(q) ||
p.author.toLowerCase().includes(q) ); if (results.length === 0) {
searchResults.innerHTML = `<p class="text-sm text-center py-4" style="color:
var(--muted);">No results found for "${searchInput.value}"</p>`; return; }
results.forEach(post => { const item = document.createElement('a'); item.href = '#';
item.className = 'flex items-center gap-4 p-4 rounded-xl transition-colors
hover:bg-[var(--surface)]'; item.innerHTML = `...`; searchResults.appendChild(item);
}); });
```

12. Step 11: Reading Progress Bar & Toast Notifications

12.1 Reading Progress Bar

The reading progress bar is a 2px tall fixed element at the very top of the viewport (z-index 10001, above the navigation). It uses the accent color and its width is dynamically updated based on the user's scroll position. The width is calculated as a percentage: $(\text{scrollTop} / (\text{documentHeight} - \text{viewportHeight})) * 100$. This creates a visual indicator of how far the user has scrolled through the page. The transition on width (0.1s linear) ensures smooth updates without jank. The bar starts at 0% width and grows to 100% as the user reaches the bottom of the page.

```
<!-- Reading Progress HTML --> <div id="reading-progress" aria-hidden="true"></div> /*
Reading Progress CSS */ #reading-progress { position: fixed; top: 0; left: 0; height:
2px; background: var(--accent); z-index: 10001; transition: width 0.1s linear; width:
0%; } /* Reading Progress JavaScript */ const progressBar =
document.getElementById('reading-progress'); window.addEventListener('scroll', () => {
const scrollTop = window.scrollTop; const docHeight =
document.documentElement.scrollHeight - window.innerHeight; const progress = docHeight
> 0 ? (scrollTop / docHeight) * 100 : 0; progressBar.style.width = progress + '%'; }, {
passive: true });
```

12.2 Toast Notifications

The toast notification system provides non-intrusive feedback for user actions like newsletter subscriptions. The toast element is fixed at the bottom-right of the screen (bottom: 2rem, right: 2rem) with a z-index of 9999. It starts hidden with `translateY(120%)` and `opacity: 0`, then slides up and fades in when the "show" class is added. The `showToast()` function sets the message text, adds the "show" class, and removes it after 3.5 seconds. The cubic-bezier (0.16, 1, 0.3, 1) transition creates a spring-like entrance animation.

```
/* Toast CSS */ .toast { position: fixed; bottom: 2rem; right: 2rem; background:
var(--surface); border: 1px solid var(--accent); color: var(--fg); padding: 1rem
1.5rem; border-radius: 12px; font-size: 0.9rem; z-index: 9999; transform:
translateY(120%); opacity: 0; transition: all 0.4s cubic-bezier(0.16, 1, 0.3, 1); }
.toast.show { transform: translateY(0); opacity: 1; } /* Toast JavaScript */ function
showToast(message) { const toast = document.getElementById('toast'); toast.textContent
= message; toast.classList.add('show'); setTimeout( () =>
toast.classList.remove('show'), 3500 ); }
```

13. Step 12: Scroll Reveal Animations

Scroll reveal is the animation system that brings content into view as the user scrolls down the page. Elements with the "reveal" class start at opacity: 0 and translateY(32px), then transition to opacity: 1 and translateY(0) when they enter the viewport. This is implemented using the Intersection Observer API, which is significantly more performant than scroll event listeners because it runs off the main thread and only fires when elements cross the viewport threshold.

```
/* Scroll Reveal CSS */ .reveal { opacity: 0; transform: translateY(32px); transition:
opacity 0.65s cubic-bezier(0.16, 1, 0.3, 1), transform 0.65s cubic-bezier(0.16, 1, 0.3,
1); } .reveal.visible { opacity: 1; transform: translateY(0); } /* Scroll Reveal
JavaScript */ function observeRevealElements() { const observer = new
IntersectionObserver( (entries) => { entries.forEach(entry => { if
(entry.isIntersecting) { entry.target.classList.add('visible');
observer.unobserve(entry.target); } }); }, { threshold: 0.08, rootMargin: '0px 0px
-40px 0px' } ); document.querySelectorAll( '.reveal:not(.visible)' ).forEach(el =>
observer.observe(el)); } observeRevealElements();
```

The Intersection Observer is configured with a threshold of 0.08 (8% visibility triggers the animation) and a rootMargin of "0px 0px -40px 0px" (effectively shrinking the trigger zone by 40px at the bottom, so elements need to be slightly more visible before animating). Once an element becomes visible, it receives the "visible" class and is immediately unobserved, preventing re-triggering. The 0.65s duration with the cubic-bezier(0.16, 1, 0.3, 1) curve creates a fast start with gentle deceleration, feeling responsive yet elegant. Elements can set custom transition-delay via inline styles (e.g., the hero right column uses 0.15s) for staggered effects.

14. Step 13: Responsive Design & Accessibility

14.1 Responsive Breakpoints

PIXO uses Tailwind CSS's mobile-first responsive system with three key breakpoints: sm (640px), md (768px), and lg (1024px). The design philosophy is to stack elements vertically on mobile and transition to multi-column layouts on larger screens. The navigation hides desktop links below md and shows a hamburger menu instead. The hero section stacks its two columns below lg. The blog post grid goes from 1 column (default) to 2 columns (md) to 3 columns (lg). The footer goes from 1 column to 2 (sm) to 4 (lg). All touch targets are at least 44px for mobile usability.

Component	Mobile (<768px)	Tablet (768-1023px)	Desktop (1024px+)
Navigation	Hamburger + overlay	Full nav bar	Full nav bar
Hero	Stacked (content, then image)		2-column grid
Post Grid	1 column	2 columns	3 columns
Editor's Pick	Stacked (image, then content)		5-column (2+3)
Newsletter Form	Stacked (input, then button)	Row	Row
Footer	1 column	2 columns	4 columns

14.2 Accessibility Features

The PIXO interface incorporates several accessibility best practices. All interactive elements have appropriate ARIA labels (aria-label on buttons, aria-live="polite" on the toast, role="dialog" on overlays). The navigation uses role="navigation" with aria-label="Main navigation". Images have descriptive alt text. The search overlay is keyboard-navigable with ESC to close and Ctrl+K to open. Focus management ensures the search input receives focus when the overlay opens. Color contrast ratios meet WCAG guidelines: the accent color (#e8a023) on the dark background (#0a0a0a) achieves a contrast ratio of approximately 8.5:1 for large text, well above the 4.5:1 minimum.

14.3 Reduced Motion Support

Users who prefer reduced motion (set via OS accessibility settings) are respected through a prefers-reduced-motion media query that effectively disables all animations and transitions. This is critical for users with vestibular disorders or motion sensitivity. The reveal class elements are set to opacity: 1 and transform: none immediately, ensuring content is always visible without animation. All other animations are reduced to near-zero duration (0.01ms) with a single iteration, effectively disabling them while maintaining the CSS structure.

```
@media (prefers-reduced-motion: reduce) { *, *::before, *::after { animation-duration: 0.01ms !important; animation-iteration-count: 1 !important; transition-duration: 0.01ms !important; } .reveal { opacity: 1; transform: none; } }
```

14.4 Key Design Decisions Summary

Decision	Choice	Rationale
Color system	CSS custom properties	Single source of truth, easy theming
Layout	CSS Grid + Flexbox	Grid for 2D layouts, Flex for 1D alignment
Animations	CSS transitions + IntersectionObserver	GPU-accelerated, off-main-thread
Font loading	Google Fonts + font-display: swap	No FOIT, graceful fallback
Icons	Font Awesome CDN	Consistent style, easy to use
Styling approach	Tailwind utilities + custom CSS	Rapid prototyping + fine control
State management	Vanilla JS + DOM	No framework overhead, simple to understand
Scroll tracking	passive event listeners	Better scroll performance
Mobile menu	Full-screen overlay	Maximum touch target area
Search	Client-side filtering	No server needed, instant results